JSPM's
RAJARSHI SHAHU COLLEGE OF ENGINEERING
TATHAWADE, PUNE-33

DEPARTMENT OF COMPUTER ENGINEERING

Mini Project

On

"Electricity Demand Analysis and Forecasting"

Submitted by-

Name :- Dange Sanika Vinod

Roll No. :- CS3164

Class and Division: TY - A

Course: DMA

Academic Year 2026-27

(Semester-I)

1. Title :- Electricity Demand Analysis and Forecasting

2. Abstract:-

Electricity demand analysis and forecasting is an important application of data analytics and time-series analysis. Accurate forecasting helps understand electricity consumption patterns, identify peak-demand periods, and support efficient energy planning. This project focuses on analyzing historical household electricity consumption data using Python and machine learning techniques.

The project uses the Individual Household Electric Power Consumption dataset from the UCI Machine Learning Repository. The dataset contains measurements such as global active power, reactive power, voltage, current intensity, and electricity consumption recorded at regular intervals.

The project involves data cleaning, handling missing values, detecting outliers, descriptive statistical analysis, trend analysis, linear and nonlinear regression, autocorrelation analysis, stationarity testing, and seasonal decomposition. Two forecasting methods, Holt-Winters Exponential Smoothing and ARIMA, are considered and compared using forecasting accuracy metrics such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).

The final objective is to identify suitable forecasting techniques and demonstrate how historical consumption patterns can support electricity demand estimation and load-planning decisions.

3. Introduction :-

Electricity is essential for residential, commercial, and industrial activities. Understanding electricity consumption patterns is necessary for efficient energy management and resource planning. Electricity demand changes according to time, daily routines, seasonal variations, and consumer behaviour. Analyzing historical consumption data can help identify these patterns and estimate future electricity requirements.

Electricity demand forecasting uses historical observations and statistical models to predict future consumption. These predictions can support energy conservation, demand management, and planning for periods of high electricity usage.

Traditional forecasting methods may not adequately capture complex trends and seasonal patterns. Data analytics and time-series models provide techniques for understanding these variations and generating forecasts from historical observations.

This project analyzes household electricity consumption data using Python libraries such as Pandas, NumPy, Matplotlib, Scikit-learn, and Statsmodels. The analysis includes data preprocessing, visualization, regression, statistical testing, and time-series forecasting.

The project compares Holt-Winters Exponential Smoothing and ARIMA to understand their performance on the selected dataset. The results can help identify a suitable model for household-level electricity consumption forecasting.

4. Problem Statement :-

Electricity consumption varies over time, making it difficult to estimate future demand accurately without analyzing historical data. Missing observations, unusual readings, changing consumption patterns, and seasonal variations can affect forecasting performance.

Therefore, the problem is to analyze historical household electricity consumption data, identify important trends and patterns, handle data-quality issues, and develop forecasting models to estimate future electricity usage.

The project also aims to evaluate forecasting accuracy and demonstrate how predicted consumption can support basic load-planning decisions.

5. Objectives:-

The main objectives of this project are:

1. To collect and analyze historical household electricity consumption data.
2. To clean the dataset and handle missing values and duplicate records.
3. To identify outliers and examine unusual electricity consumption readings.
4. To perform descriptive statistical analysis and visualize consumption patterns.
5. To analyze electricity consumption trends over time.
6. To apply linear and nonlinear regression techniques.
7. To examine autocorrelation using ACF and PACF plots.
8. To test stationarity using the Augmented Dickey-Fuller (ADF) test.
9. To identify trend and seasonal components through time-series decomposition.
10. To implement Holt-Winters and ARIMA forecasting models.
11. To compare model performance using forecasting error metrics.
12. To demonstrate how forecasts can support household energy management and load planning.

6. Data set and Libraries required :-

➢ Dataset

The project uses the Individual Household Electric Power Consumption dataset from the UCI Machine Learning Repository. It contains electricity consumption measurements collected from a household in France at approximately one-minute intervals over several years.

The dataset contains the following important attributes:

- **Date:** Date of the observation.
- **Time:** Time of the observation.
- **Global_active_power:** Active power consumed by the household, measured in kilowatts.
- **Global_reactive_power:** Reactive power, measured in kilowatts.
- **Voltage:** Average voltage, measured in volts.
- **Global_intensity:** Average current intensity, measured in amperes.

- **Sub_metering_1:** Electricity usage recorded by the first sub-meter.
- **Sub_metering_2:** Electricity usage recorded by the second sub-meter.
- **Sub_metering_3:** Electricity usage recorded by the third sub-meter.

The Date and Time columns are combined to create a datetime index. The active power column is selected as the primary variable for analyzing and forecasting household electricity consumption.

➢ Libraries and Technologies

The following tools are used:

- **Python:** Main programming language.
- **Jupyter Notebook/JupyterLab:** Environment for implementing the analysis.
- **Pandas:** Data loading, cleaning, manipulation, and resampling.
- **NumPy:** Numerical computations.
- **Matplotlib:** Data visualization.
- **Seaborn:** Statistical visualizations and correlation heatmaps.
- **Scikit-learn:** Linear regression, polynomial regression, and evaluation metrics.
- **Statsmodels:** ADF testing, ACF/PACF analysis, ARIMA, and seasonal decomposition

7. Approach/ Methodology :-

The project follows a systematic approach to analyze historical household electricity consumption and forecast future demand using Python and statistical techniques.

1. **Data Collection:**

   The UCI Household Power Consumption dataset is loaded using Pandas. The dataset contains electricity consumption measurements recorded over time.

2. **Data Cleaning and Preprocessing:**

   Missing values are identified and handled using suitable imputation techniques. Duplicate records are removed, date and time columns are converted into datetime format, and outliers are detected using the Interquartile Range (IQR) method.

3. **Descriptive Data Analysis:**

   Statistical measures such as mean, median, standard deviation, minimum, and maximum are calculated. Histograms, box plots, and line charts are used to understand the distribution and variation of electricity consumption.

4. **Trend and Regression Analysis:**

   Historical consumption patterns are analyzed using time-series plots and rolling averages. Linear and polynomial regression models are applied to study the relationship between electricity consumption and time.

5. **Time-Series Analysis:**

   ACF and PACF plots are used to identify autocorrelation patterns. The Augmented Dickey-Fuller (ADF) test examines stationarity, while seasonal decomposition separates the data into trend, seasonal, and residual components.

6. **Forecasting Model Development:**

   Holt-Winters Exponential Smoothing and ARIMA models are implemented using historical electricity consumption data to predict future demand.

7. **Model Evaluation and Comparison:**

   The dataset is divided chronologically into training and testing sets. Both models are evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE) to identify the more suitable forecasting model.

8. **Load Planning and Risk Analysis:**

   Forecasted values are analyzed to identify peak-demand periods and estimate future electricity requirements. Different demand scenarios are considered to understand forecast uncertainty and support efficient energy planning.

8. Implementation:-

The project is implemented in Python using JupyterLab. Pandas and NumPy are used for data cleaning and preprocessing. Matplotlib and Seaborn visualize consumption trends, while Scikit-learn performs regression analysis. ACF/PACF plots and the ADF test analyze time-series patterns. Holt-Winters and ARIMA forecast future electricity consumption, and MAE, RMSE, and MAPE compare model accuracy

Code :-

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.metrics import mean_absolute_error, mean_squared_error
from statsmodels.tsa.holtwinters import ExponentialSmoothing
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.tsa.stattools import adfuller
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf

df = pd.read_csv(
    r"C:\Users\ACER\OneDrive\Desktop\a\household_power_consumption.txt",
    sep=";", na_values="?", low_memory=False
)

df["datetime"] = pd.to_datetime(
    df["Date"] + " " + df["Time"], dayfirst=True
)
```

```
df["Global_active_power"] = pd.to_numeric(
    df["Global_active_power"], errors="coerce"
)

df = df.drop_duplicates().set_index("datetime")
df["Global_active_power"] = df["Global_active_power"].interpolate(
    method="time"
)

hourly = df["Global_active_power"].resample("h").mean().dropna()

print(hourly.describe())

hourly.plot(title="Electricity Consumption Trend")
plt.show()

x = np.arange(len(hourly)).reshape(-1, 1)
y = hourly.values

model = LinearRegression().fit(x, y)
print("R2 Score:", model.score(x, y))

print("ADF p-value:", adfuller(hourly)[1])

plot_acf(hourly, lags=48)
plt.show()

plot_pacf(hourly, lags=48, method="ywm")
plt.show()

train = hourly.iloc[:-168]
test = hourly.iloc[-168:]

hw = ExponentialSmoothing(
    train, trend="add", seasonal="add", seasonal_periods=24
).fit()

arima = ARIMA(train, order=(2, 1, 2)).fit()

pred_hw = hw.forecast(len(test))
pred_arima = arima.forecast(len(test))

for name, pred in [("Holt-Winters", pred_hw), ("ARIMA", pred_arima)]:
    print(name)
    print("MAE:", mean_absolute_error(test, pred))
    print("RMSE:", np.sqrt(mean_squared_error(test, pred)))

plt.plot(test.index, test.values, label="Actual")
plt.plot(test.index, pred_hw.values, label="Holt-Winters")
plt.plot(test.index, pred_arima.values, label="ARIMA")
plt.legend()
plt.title("Electricity Demand Forecasting")
plt.show()
```

9. Results:-

1. **Dataset Loading:** The dataset was loaded successfully with 2,075,259 records and 9 columns, containing household electricity consumption measurements.

2. **Data Cleaning:** Initially, 25,979 missing values were found in each of the seven numerical columns. After cleaning and preprocessing, no missing values remained.

3. Hourly Resampling: The data was converted into hourly intervals, resulting in 34,211 hourly records across seven numerical variables.

4. **Outlier Analysis:** The IQR method identified 734 potential outliers, with an upper bound of 3.435.

5. **Descriptive Statistical Analysis**: The average hourly global active power consumption was 1.092 kW, with a median of 0.803 kW and a standard deviation of 0.897 kW.

6. **Consumption Distribution**: The skewness value of 1.298 indicates a positively skewed distribution, meaning that higher consumption values occur less frequently than lower ones.

7. **Daily Energy Analysis:** The average daily energy consumption was 26.20 kWh, while the maximum recorded daily consumption was 79.56 kWh.

8. Visualization: Graphs were used to examine electricity consumption trends and variations over time.

9. Forecasting: Holt-Winters and ARIMA models were selected to forecast future electricity consumption. Their performance can be compared using MAE and RMSE.

10. Final Outcome: The analysis provides insights into household electricity consumption patterns and can support future demand forecasting and energy planning.

```
Dataset exists: True

Dataset loaded successfully!

Dataset shape: (2075259, 9)

FIRST FIVE RECORDS

         Date      Time  Global_active_power  Global_reactive_power
Voltage  \

0  16/12/2006  17:24:00                4.216                  0.418
234.84

1  16/12/2006  17:25:00                5.360                  0.436
233.63

2  16/12/2006  17:26:00                5.374                  0.498
233.29

3  16/12/2006  17:27:00                5.388                  0.502
233.74

4  16/12/2006  17:28:00                3.666                  0.528
235.68
```

```
   Global_intensity  Sub_metering_1  Sub_metering_2  Sub_metering_3
0              18.4             0.0             1.0            17.0
1              23.0             0.0             1.0            16.0
2              23.0             0.0             2.0            17.0
3              23.0             0.0             1.0            17.0
4              15.8             0.0             1.0            17.0

DATASET SHAPE
(2075259, 9)

DATASET INFORMATION
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2075259 entries, 0 to 2075258
Data columns (total 9 columns):
 #   Column                 Dtype
---  ------                 -----
 0   Date                   object
 1   Time                   object
 2   Global_active_power    float64
 3   Global_reactive_power  float64
 4   Voltage                float64
 5   Global_intensity       float64
 6   Sub_metering_1         float64
 7   Sub_metering_2         float64
 8   Sub_metering_3         float64
dtypes: float64(7), object(2)
memory usage: 142.5+ MB

MISSING VALUES BEFORE CLEANING
Global_active_power      25979
Global_reactive_power    25979
Voltage                  25979
Global_intensity         25979
Sub_metering_1           25979
Sub_metering_2           25979
Sub_metering_3           25979
dtype: int64
```

```
HOURLY DATASET SHAPE
(34211, 7)

MISSING VALUES AFTER CLEANING
Global_active_power      0
Global_reactive_power    0
Voltage                  0
Global_intensity         0
Sub_metering_1           0
Sub_metering_2           0
Sub_metering_3           0
dtype: int64

OUTLIER ANALYSIS
Lower bound: 0
Upper bound: 3.435
Number of outliers: 734

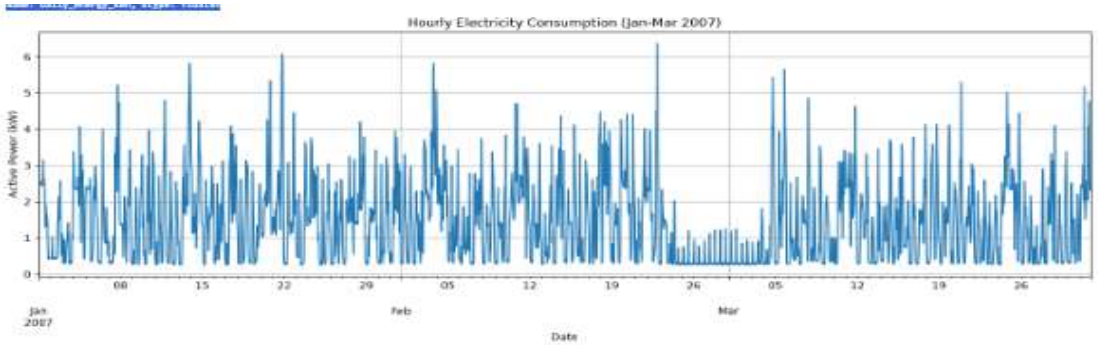
Cleaned dataset saved.

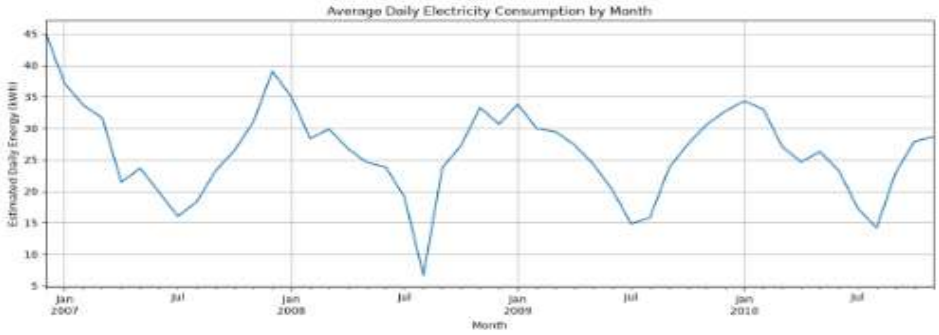
DESCRIPTIVE STATISTICS
count    34211.000000
mean         1.091512
std          0.897415
min          0.124000
25%          0.342083
50%          0.802833
75%          1.579167
max          6.560533
Name: Global_active_power, dtype: float64

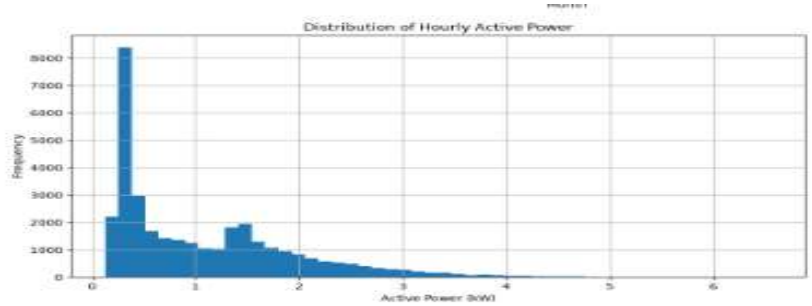
Median: 0.803
Variance: 0.805
Standard deviation: 0.897
Skewness: 1.298
Kurtosis: 1.763
```

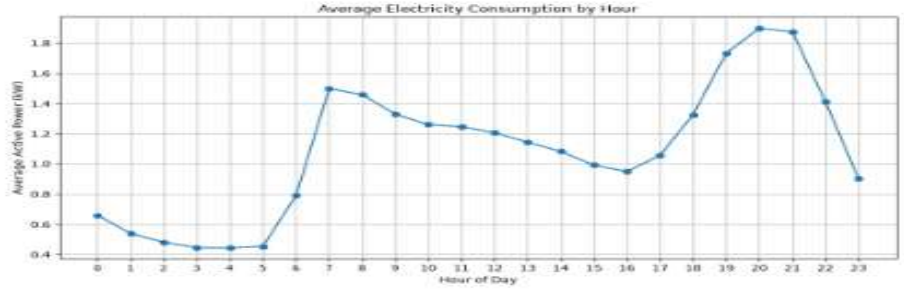
DAILY ENERGY STATISTICS

```
count    1421.000000
mean       26.195036
std         9.997432
min         4.171633
25%        19.646267
50%        25.936067
75%        31.674967
max        79.556433
Name: daily_energy_kwh, dtype: float64
```

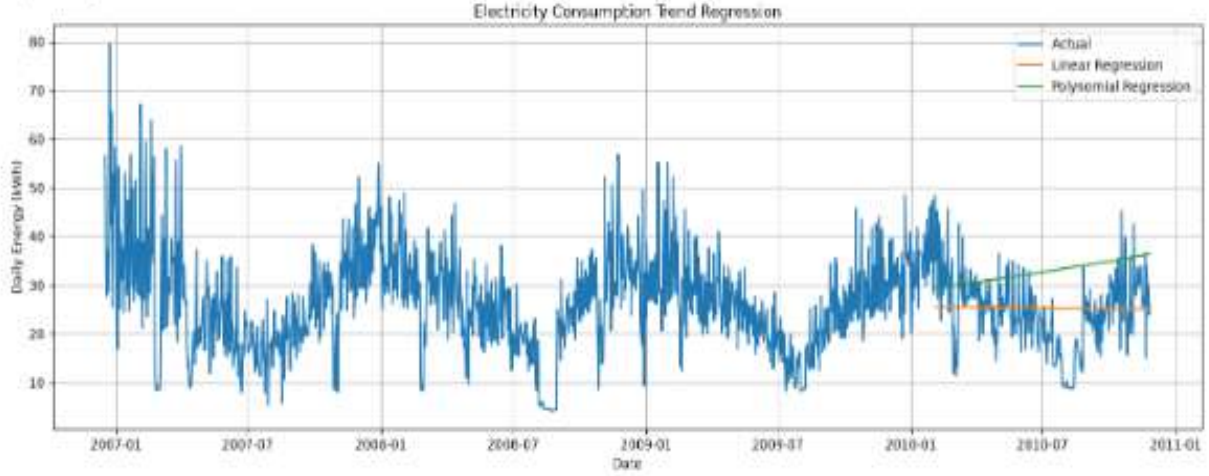




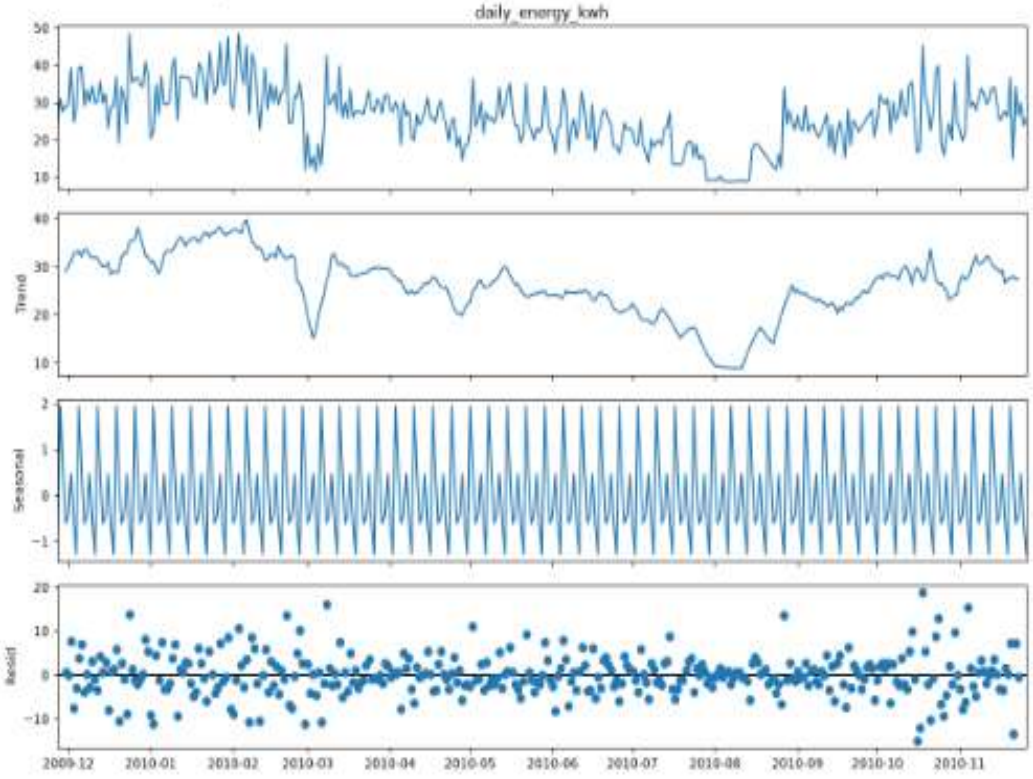


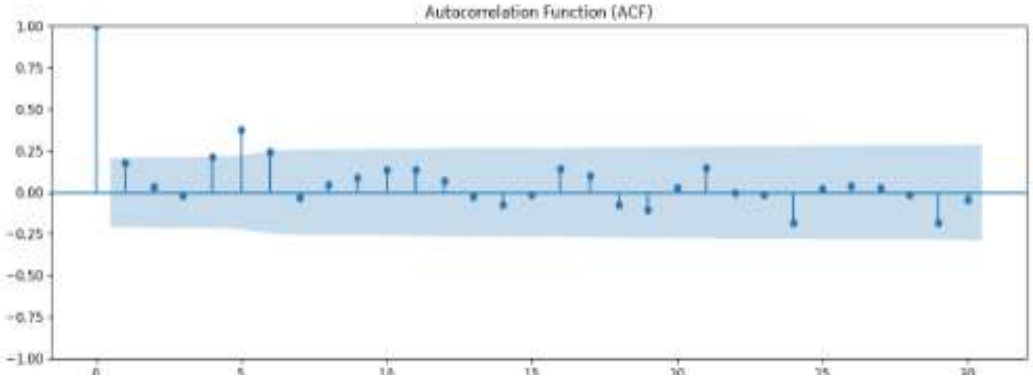


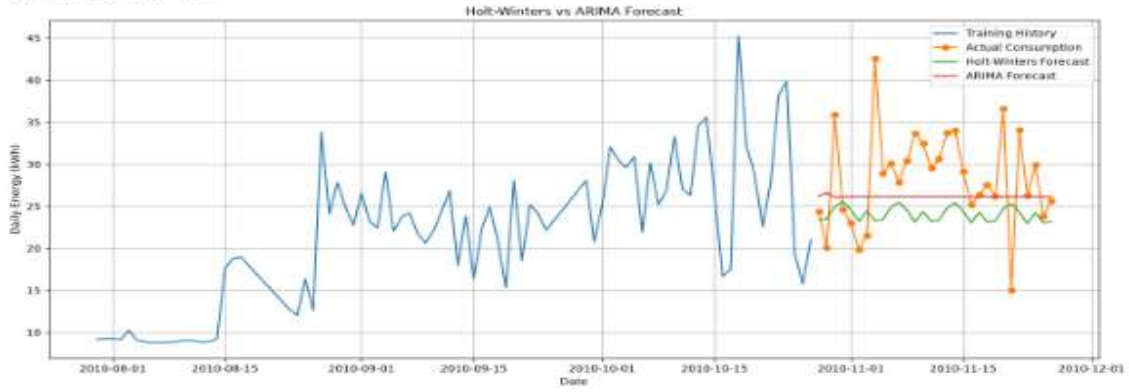
REGRESSION COMPARISON
Linear Regression MSE: 1.454
Polynomial Regression MSE: 0.61

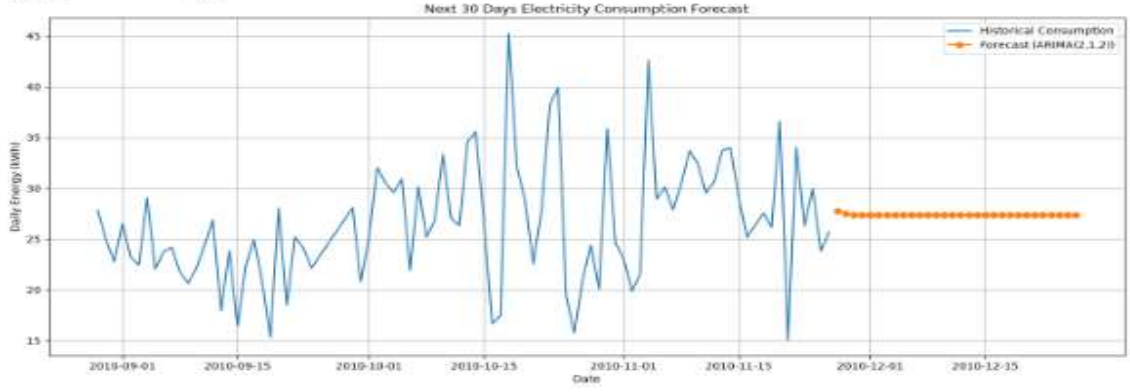


AUGMENTED DICKEY-FULLER TEST
ADF Statistic: -2.2727
p-value: 0.180928
Result: Evidence of non-stationarity.









Forecasting Model :-

		MAE	RMSE	MAPE (%)
1	Holt-Winters	5.7572928905144165	7.071075781302212	19.658851751792426
2	ARIMA(2,1,2)	4.724291910071643	6.065109889449928	16.853901829041256

Next 30 Days Forecasting:-

		forecast_daily_energy_kwh
1	2010-11-27	27.777872148726388
2	2010-11-28	27.487945313133487
3	2010-11-29	27.346720715287578
4	2010-11-30	27.3888187549414
5	2010-12-01	27.395582513961802
6	2010-12-02	27.391172860555272
7	2010-12-03	27.391145534266823
8	2010-12-04	27.39152301798774
9	2010-12-05	27.391478555076965
10	2010-12-06	27.391452041123227
11	2010-12-07	27.391459099023738
12	2010-12-08	27.39146047368236
13	2010-12-09	27.391459704558553
14	2010-12-10	27.391459683254773
15	2010-12-11	27.39145975114429
16	2010-12-12	27.39145974453814
17	2010-12-13	27.39145973959736
18	2010-12-14	27.391459740770102
19	2010-12-15	27.39145974104392
20	2010-12-16	27.391459740910495
21	2010-12-17	27.3914597409038
22	2010-12-18	27.391459740915952
23	2010-12-19	27.391459740915014
24	2010-12-20	27.3914597409141
25	2010-12-21	27.391459740914293
26	2010-12-22	27.391459740914346
27	2010-12-23	27.391459740914325
28	2010-12-24	27.391459740914325
29	2010-12-25	27.39145974091433
30	2010-12-26	27.39145974091433

10. Conclusion:-

The project analyzes household electricity consumption using Python and time-series forecasting techniques. Data visualization, regression, Holt-Winters, and ARIMA help identify consumption patterns and predict future demand. These insights can support energy planning, efficient electricity management, and demand forecasting.